Department of Computer Science
CS 298/299 Proposal
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GenderStego: A Novel Framework for Gait-Based Gender Recognition and Video Steganography

Sohan Vallapureddy

Advisor
Dr. Nada Attar, nada.attar@sjsu.edu

Committee Members
Dr. Amith Kamath Belman, amith.kamathbelman@sjsu.edu
Dr. Noha Elfiky, nme5@stmarys-ca.edu

Introduction

GenderStego is an innovative framework that combines gait-based gender recognition with advanced video steganography techniques. This project aims to develop a comprehensive system that classifies gender from human gait patterns while concurrently providing a secure method for embedding and storing sensitive video data.

Keywords: *Computer Vision, Pose Estimation, Gait-based Gender Recognition, DCT-based Steganography, Biometric Privacy, Ethical AI*

Problem Definition and Motivation

GenderStego addresses critical challenges in biometrics, data security, and ethics by combining non-invasive gait-based gender recognition with advanced video steganography. The project offers a novel solution for gender classification from a distance while securely storing sensitive video content, addressing the limitations of traditional methods and existing steganography techniques. By incorporating privacy-preserving techniques, GenderStego tackles ethical concerns surrounding biometric data collection and storage. This approach is particularly relevant in the current landscape of increased surveillance and data privacy issues, offering applications in security and forensics. The project contributes to both the research community and the broader public by balancing technological advancement with responsible data practices.

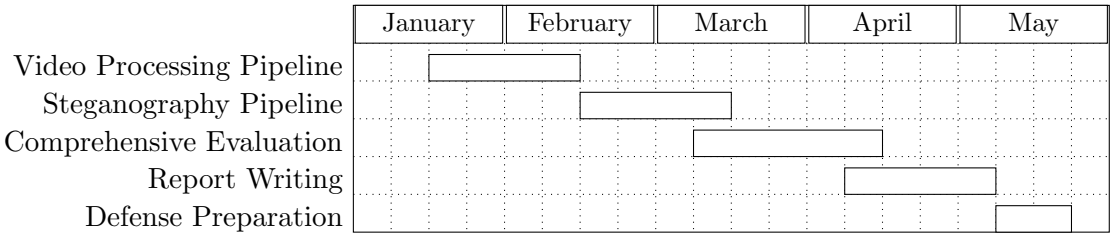
Results Achieved in CS 297

- Gait-Based Gender Recognition Module:
  - Implement a pose estimation algorithm using MediaPipe to extract skeletal keypoints from video frames.
  - Design and train a neural network model using the OU-ISIR database, with pose features as input and gender as the target variable.

Expected Deliverables in CS 298

- Detect and remove humans to generate clean videos using inpainting techniques.
- Secure Steganography Pipeline:
  - Use DCT-based steganography for embedding original video into inpainted video
  - Implement AES-256 encryption for embedded data and HMAC for authentication
- Comprehensive Evaluation:
  - Evaluate metrics for gender classification using accuracy, precision, recall, and F1-score on a held-out test set
  - Quantitative metrics for steganography performance (capacity, PSNR, extraction speed)
  - Ethical Evaluation by performing bias testing across diverse demographics (age, ethnicity, body types)
- Document ReadMe containing detailed instructions on how to run the code and APIs if any

Timeline and Milestones for CS 298



References

[1] A. Elhadad, S. Hamad, A. Khalifa, and A. Ghareeb, "High capacity information hiding for privacy protection in digital video files," *Neural Computing and Applications*, vol. 28, 2017.

[2] C.-E. Sung and N. Attar, "Gender classification accuracy via two-dimensional body joints using convolutional neural networks," in *2023 IEEE International Conference on Big Data (Big Data)*, pp. 2229–2233, 2023.

[3] A. Elhadad, S. Hamad, A. Khalifa, and H. Abulkasim, "A steganography approach for hiding privacy in video surveillance systems," in *Digital Media Steganography: Principles, Algorithms, and Advances*, pp. 165–187, Elsevier, 2020.

[4] E. Salcedo, "Computer vision-based gait recognition on the edge: A survey on feature representations, models, and architectures," *Journal of Imaging*, vol. 10, no. 12, 2024.

[5] M. Azhar, S. Ullah, K. Ullah, I. Syed, and J. Choi, "A gait-based real-time gender classification system using whole body joints," *Sensors*, vol. 22, p. 9113, Nov 2022.

[6] A. Cosma and I. E. Radoi, "Wildgait: Learning gait representations from raw surveillance streams," *Sensors*, vol. 21, no. 24, 2021.

[7] A. Cosma and E. Radoi, "Learning gait representations with noisy multi-task learning," *Sensors*, vol. 22, no. 18, 2022.